

Customer Subscription Prediction for Bank Marketing: A Tradeoff Between Recall and Targeting Efficiency

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Github Repository: <https://github.com/Cassielu77/Bank-Deposit-Prediction>

Abstract—Direct marketing campaigns are widely used in the banking industry to promote financial products such as term deposits. However, identifying customers who are likely to subscribe remains challenging because customer response behavior is highly imbalanced and influenced by multiple demographic, financial, and campaign-related factors. Inefficient targeting can lead to unnecessary operational costs and low marketing conversion rates.

This project aims to predict whether a bank customer will subscribe to a term deposit using machine learning classification models. Using the UCI Bank Marketing dataset, we developed an end-to-end machine learning pipeline including data exploration, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation.

Two supervised classification models, Logistic Regression and Decision Tree, were implemented entirely from scratch using NumPy. The models were evaluated using Accuracy, Precision, Recall, F1-score, F2-score, AUC-ROC, AUC-PR, and ALIFT. Because the dataset is highly imbalanced, the project focused particularly on Recall, F2-score, and ranking-based metrics that better reflect marketing objectives.

The experimental results show that both models learned meaningful predictive patterns from customer and campaign data, but they exhibited different strengths. Logistic Regression achieved higher Recall and identified a larger proportion of potential subscribers, making it more suitable for aggressive lead capture strategies. In contrast, the Decision Tree achieved stronger overall Precision, F1-score, F2-score, AUC-ROC, AUC-PR, and ALIFT, providing a more balanced tradeoff between subscriber identification and operational efficiency. Both models consistently identified previous campaign success and customer contact information as the strongest predictors of subscription behavior.

The findings demonstrate that model selection in bank marketing prediction should depend on business objectives rather than a single evaluation metric. The project also highlights how interpretable machine learning models can support data-driven customer targeting strategies while improving marketing

efficiency and reducing unnecessary outreach costs.

Index Terms—machine learning, bank marketing, binary classification, logistic regression, decision tree, Streamlit, interpretable AI

I. INTRODUCTION

Direct marketing is an important strategy for banks to promote financial products such as term deposits. However, traditional marketing campaigns often contact a large number of customers without knowing whether they are likely to subscribe. This creates a practical business problem: banks may spend time and labor on customers who are unlikely to respond, while also missing customers who have a higher probability of subscription. Therefore, predicting customer subscription behavior before outreach can help banks improve marketing efficiency, reduce unnecessary customer contact, and allocate campaign resources more effectively.

Prior research has shown that machine learning can be useful for bank marketing prediction. Previous studies using the Bank Marketing dataset have applied models such as Logistic Regression, Decision Trees, Support Vector Machines, Neural Networks, and other classification methods to predict whether a customer will subscribe to a term deposit [1], [2]. These studies demonstrate that demographic, financial, and campaign-related features can provide meaningful signals for customer response prediction. However, many prior approaches focus mainly on maximizing predictive performance or comparing advanced algorithms, while less attention is given to interpretability, class imbalance, and realistic deployment for pre-contact marketing decisions. Since customer response data is highly imbalanced, standard classifiers may become biased toward the majority class, and evaluation based only on Accuracy can be misleading [6].

This project addresses these gaps by building an end-to-end machine learning pipeline that emphasizes both predictive performance and practical usefulness. Because the dataset is highly imbalanced, where only a small share of customers subscribe, Accuracy alone is not enough to evaluate model quality. A model that predicts most customers as non-subscribers could still achieve high Accuracy but fail to identify the customers most valuable to the bank. For this reason, our project focuses on evaluation metrics that better match the business objective, including Precision, Recall, F1-score, F2-score, Area Under the Receiver Operating Characteristic Curve (AUC-ROC), Area Under the Precision-Recall Curve (AUC-PR), and ALIFT.

Our solution compares two supervised machine learning models implemented from scratch using NumPy: Logistic Regression and Decision Tree. Logistic Regression is useful because it provides interpretable coefficients and stable probability estimates, making it easier to understand how each feature contributes to the predicted subscription probability. Decision Tree is also interpretable, but it can capture nonlinear relationships and threshold-based rules among customer and campaign features. Comparing these two models allows us to study the tradeoff between recall-oriented lead capture and more efficient customer targeting.

In addition to offline model training and evaluation, we developed a Streamlit web application to make the prediction system usable for non-technical users. The target users are bank marketing teams, campaign managers, and outbound calling staff. Through the application, users can enter customer and campaign information, adjust the prediction threshold, and receive a predicted subscription probability and binary recommendation. This connects the trained models to a practical decision-support tool that can help marketing teams prioritize customers before making contact.

The broader impact of this project is that machine learning can support more data-driven and efficient financial marketing. By ranking customers based on predicted subscription probability, banks can reduce wasted outreach, improve campaign planning, and focus attention on customers who are more likely to benefit from the offered product. At the same time, because customer targeting can influence who receives financial product offers, the project also highlights the need for interpretable models, careful evaluation, and future fairness analysis before real-world deployment.

II. BACKGROUND

Customer response prediction in bank direct marketing has been widely studied as an important binary classification problem in financial analytics. In traditional telemarketing campaigns, banks often contact a large number of customers without knowing in advance who is likely to subscribe to a term deposit. This leads to high operational cost and inefficient use of marketing resources. Early studies showed that customer demographic, financial, and campaign-related variables can be used to support prediction before outreach decisions are made. In particular, Moro et al. demonstrated that multiple

classification methods, including Logistic Regression, Decision Trees, Support Vector Machines, and Neural Networks, can be applied effectively to predict term deposit subscription. Their work also helped establish the Bank Marketing dataset as a benchmark for this research area [1]. Earlier work by Moro, Laureano, and Cortez further showed that data mining methods can be integrated into direct marketing analysis through the CRISP-DM framework, highlighting the practical relevance of predictive modeling in campaign planning [2].

As research progressed, the broader financial prediction literature gradually moved beyond traditional single-model methods toward more advanced ensemble approaches. Lessmann et al. found that tree-based ensemble models could outperform more conventional methods in financial classification tasks such as credit scoring, suggesting a broader methodological shift toward models that can better capture nonlinear relationships and interactions among predictors [3]. This trend was reinforced by the development of boosting methods such as XGBoost, which provided a scalable and effective tree-boosting framework for structured tabular data [4]. In a similar direction, stacking methods were proposed to combine predictions from multiple base learners into a stronger final predictor [5]. Taken together, these studies show that the field has increasingly emphasized predictive performance through more complex ensemble systems.

However, several important challenges remain in marketing response prediction. One major issue is class imbalance. In direct marketing datasets, only a small proportion of customers typically subscribe to the offered product, while the majority do not. He and Garcia identified imbalanced learning as a fundamental machine learning problem, showing that standard classifiers may become biased toward the majority class and that Accuracy alone may therefore be misleading [6]. In the context of bank marketing, this means model evaluation should also consider Precision, Recall, F1-score, and ROC-AUC in order to better measure performance on the minority class of real business interest. In direct marketing contexts, however, these metrics remain limited because they depend on a fixed decision threshold and do not reflect how well a model prioritises likely subscribers within a ranked contact list. To address this, the Cumulative Gains Curve has been widely adopted in campaign evaluation to show what proportion of true subscribers can be captured by contacting only the top-scoring fraction of customers, directly translating model performance into budget planning decisions. Building on this, Piatetsky-Shapiro and Steingold formalised the area under this curve as a single ranking-quality metric, allowing models to be compared without relying on any specific threshold [7].

Another key issue is the trade-off between predictive performance and interpretability. Although ensemble models often achieve stronger classification results, they may be harder to interpret and explain in practice. In financial decision-support settings, marketing managers usually need transparent models that allow them to understand why a customer is predicted to subscribe or not subscribe before using the results to guide campaign strategy. A further concern is deployment realism.

In the Bank Marketing dataset, some variables, such as call duration, are highly predictive but are only available after the customer interaction has already taken place, which can introduce data leakage if used in a real pre-contact prediction scenario [1]. In addition, much of the prior literature focuses mainly on comparing algorithms rather than building deployable systems that connect preprocessing, training, evaluation, and real-time inference.

Based on this literature, our project places greater emphasis on interpretability and practical deployment. Instead of focusing only on maximizing predictive accuracy through complex ensemble methods, we use Logistic Regression and Decision Tree models within a complete end-to-end workflow. This design allows the project to connect prior research on bank marketing prediction with a more transparent and user-oriented decision-support application.

III. END-TO-END ML PIPELINE

A. Offline Model Training and Evaluation

The back-end of the web application performs the core machine learning workflow, including data exploration, preprocessing, model training, evaluation, and deployment support. It also handles the transformation of user input from the front-end into the same format used during model training, ensuring consistency between offline development and real-time inference.

1) *Data Collection, Exploration & Processing*: We used the Bank Marketing dataset from the UC Irvine Machine Learning Repository. The dataset was released in 2012 and contains 45,211 customer records collected from a banking institution's direct marketing campaigns. Each row represents one customer contact record. The dataset includes both numerical and categorical variables. Numerical variables include age, balance, day_of_week, campaign, pdays, and previous. Categorical variables include job, marital status, education, contact type, month, and previous campaign outcome. The target variable is y, which indicates whether the customer subscribed to a term deposit.

The dataset provides target labels and we converted this target into a binary label y_binary, where 1 indicates subscription and 0 indicates non-subscription. The dataset was used to train and evaluate two supervised binary classification models implemented afterwards, Logistic Regression and Decision Tree.

We first conducted exploratory data analysis to better understand the structure of the dataset and identify potentially predictive variables. A bar chart and a pie chart of the target variable revealed a strong class imbalance, with approximately 88.3% of customers belonging to the non-subscriber class and only 11.7% belonging to the subscriber class. This observation is important because it shows that accuracy alone would not be an appropriate evaluation metric for the prediction task. Moreover, we also examined numerical variables using histograms and boxplots. The balance variable showed strong right skew with several extreme values. Similar skewness patterns were observed for the campaign and the previous one. The pdays

variable contained a large number of -1 values, indicating customers who had not been contacted previously. These patterns motivated later preprocessing and feature engineering decisions.

For categorical variables, we visualized distributions for features such as job, marital status, contact type, month, and poutcome. The analysis suggested that previous campaign outcomes and communication channel information differed substantially across customer subscription groups. These visualizations helped identify variables that were likely to contribute to model performance before training. Several preprocessing techniques were applied before model training. Missing values in categorical columns such as job, education, and contact were filled using the category unknown. Missing values in poutcome were filled with not_contacted to indicate no previous campaign history. We removed the duration feature because call duration is only known after the phone call ends. Including this feature would introduce data leakage and produce unrealistic prediction performance in a real-world deployment setting.

Next, binary categorical variables, including default, housing, and loan, were encoded as 1 and 0. For the pdays feature, we engineered two additional variables. The first variable, previous_contact, indicates whether the customer had been contacted previously. We also corrected skewed numerical variables before training. The balance feature was transformed using a shifted logarithmic transformation to reduce heavy right skew. The campaign variable was capped using the interquartile range method to reduce the influence of extreme outliers. The previous variable was transformed using log1p to stabilize variance while preserving zero values. Finally, categorical variables including job, marital, education, contact, month, and poutcome were one-hot encoded without dropping the first category. After preprocessing, the final feature matrix contained 48 input features.

2) *Methods and Model Training*: We formulated the task as a binary classification problem. Given a customer feature vector containing demographic, financial, and campaign-related information, the goal is to predict whether the customer will subscribe to a term deposit. The target label is binary, where 1 represents subscription and 0 represents non-subscription. We implemented two machine learning algorithms entirely from scratch using NumPy: Logistic Regression and Decision Tree. No machine learning training libraries such as scikit-learn were used for model implementation.

Logistic Regression models the probability of customer subscription using the sigmoid function applied to a linear combination of input features:

$$P(y = 1 | x) = \sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (1)$$

The model parameters were optimized using Binary Cross-Entropy loss:

$$L(w, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

. To address class imbalance, positive samples were assigned larger gradient weights proportional to the ratio between negative and positive samples. We also added L2 regularization to reduce overfitting by penalizing large weights:

$$L_{reg} = L + \lambda ||w||^2 \quad (3)$$

. The bias term was excluded from regularization. Optimization was performed using gradient descent with a learning rate 0.5, a regularization strength 0.01, and 3,000 iterations. Continuous variables such as age, balance, day_of_week, campaign, pdays_clean, and previous, were standardized using the training-set mean and standard deviation. Binary and one-hot encoded variables remained as 0 and 1 values. The dataset was split using stratified 80/20 train-test sampling with random seed 42. A prediction threshold of 0.4 was selected to prioritize recall for the minority subscriber class while maintaining reasonable precision.

The Decision Tree classifier was implemented using recursive binary splitting with Gini impurity as the splitting criterion. At each node, the algorithm searches for the feature and threshold combination that minimizes weighted Gini impurity:

$$Gini = 1 - \sum_{k=1}^n p_k^2 \quad (4)$$

.For binary classification, this simplifies to:

$$Gini_{split} = \frac{n_{left}}{n} Gini_{left} + \frac{n_{right}}{n} Gini_{right} \quad (5)$$

where p_1 represents the proportion of positive samples within the node. Lower Gini impurity indicates better class separation. The tree incorporates class balancing through weighted samples so that minority-class customers contribute proportionally more during splitting. Predicted probabilities are computed based on weighted class frequencies within each leaf node. To avoid overfitting, the Decision Tree uses several stopping conditions, including maximum tree depth and minimum samples required for splitting. Hyperparameter tuning was conducted using grid search across different values of maximum depth and minimum split size.

We think Logistic Regression is suitable for this problem because it provides more interpretable coefficients and stable probability estimates. The model allows us to understand how each customer feature contributes to subscription probability through a linear decision boundary. A decision tree is also interpretable while being capable of capturing nonlinear interactions and threshold-based relationships among variables. This flexibility is important because customer response behavior may depend on combinations of previous campaign outcome, contact method, timing, and financial characteristics. The implementation of these two models provides a useful comparison between a linear probabilistic classifier and a nonlinear rule-based classifier.

Both models receive the same preprocessed customer feature matrix containing transformed numerical variables, binary encoded variables, and one-hot encoded categorical variables.

The output of each model is a predicted probability representing the likelihood of subscription and a binary prediction obtained by applying a decision threshold to the predicted probability.

3) *Model Evaluation:* We implemented multiple evaluation metrics entirely from scratch to evaluate the performance of the term deposit prediction models. Because of the imbalance in the dataset, relying on a single metric such as Accuracy would not provide a reliable measure of model quality. Therefore, we evaluated the models using Accuracy, Precision, Recall, F1-score, F2-score, Area Under the Receiver Operating Characteristic Curve (AUC-ROC), Area Under the Precision-Recall Curve (AUC-PR), and ALIFT. Accuracy measures the proportion of correctly classified observations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

Although Accuracy is intuitive, it is misleading for highly imbalanced datasets. Since approximately 88.3% of customers in the dataset are non-subscribers, a trivial classifier predicting every customer as a non-subscriber would still achieve high Accuracy while failing completely at identifying potential subscribers. Precision measures the proportion of predicted positive cases that are actually positive:

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

. Precision is important from a business perspective because low Precision would lead to unnecessary marketing calls and wasted operational resources. Recall measures the proportion of actual subscribers correctly identified by the model:

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

. We think Recall is an important indicator here because there will be a lost business opportunity if we fail to identify a potential subscriber. In direct marketing campaigns, missing a potential customer is often more costly than contacting a non-subscriber.

Furthermore, F1-score balances Precision and Recall through their harmonic mean:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (9)$$

. We also used F2-score, which places greater emphasis on Recall:

$$F2 = \frac{5 \cdot Precision \cdot Recall}{4 \cdot Precision + Recall}$$

. The F2-score was treated as one of the primary evaluation metrics because the project objective prioritizes identifying as many potential subscribers as possible, even if this results in additional false positives. Weighting Recall more heavily than Precision aligns the model evaluation process with the practical business objective of maximizing lead capture.

AUC-ROC evaluates the model's ability to distinguish between positive and negative classes across all classification thresholds. AUC-PR measures the tradeoff between Precision

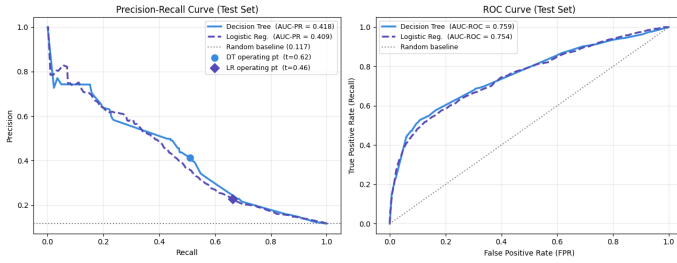


Fig. 1. The relationship between threshold selection and model performance on the validation set

and Recall across thresholds and is especially informative for imbalanced datasets because it focuses more directly on minority-class performance. We also evaluated ALIFT, which measures how effectively the model concentrates true subscribers within the highest-ranked prediction groups. This metric is useful in marketing scenarios where customer outreach is prioritized according to predicted probability scores.

The dataset was divided using stratified 64/16/20 train-validation-test splitting with random seed 42. Stratified sampling preserves the original positive-class ratio across both datasets and ensures stable evaluation under class imbalance. The training set was used to optimize model parameters, including Logistic Regression weights and Decision Tree split structures. The test set was held out entirely during training and used only for final evaluation to provide an unbiased estimate of model generalization performance on unseen customer data.

Feature standardization was applied using only training-set statistics for our logistic regression. The mean and standard deviation computed from the training set were applied to both training and testing data to avoid data leakage.

4) *Results:* For Logistic Regression, we used a learning rate of 0.5, L2 regularization strength of 0.01, and 3,000 gradient descent iterations after verifying stable convergence behavior during training. A prediction threshold of 0.46 was selected because it produced the highest F2-score while maintaining balanced Precision and Recall. For the Decision Tree, the selected operating threshold was 0.62. Hyperparameter tuning for the Decision Tree was conducted using grid search over different values of maximum depth and minimum samples per split.

Figure 1 shows the relationship between threshold selection and model performance on the validation set. For both models, F2-score peaks at a lower threshold than F1-score because F2 places greater emphasis on Recall. The Logistic Regression model achieved its best F2-score near threshold 0.46, while the Decision Tree achieved its best F2-score near threshold 0.62. This threshold analysis demonstrates how changing the decision threshold directly affects the tradeoff between Precision and Recall.

The confusion matrices in Figure 2 provide a direct comparison of prediction behavior on the test set. Logistic Regression identified more true subscribers, producing 702 true

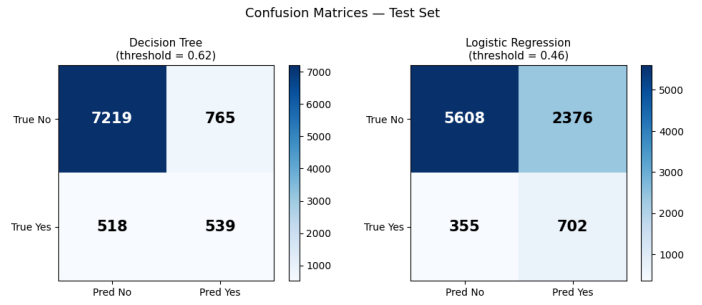


Fig. 2. Confusion Matrix

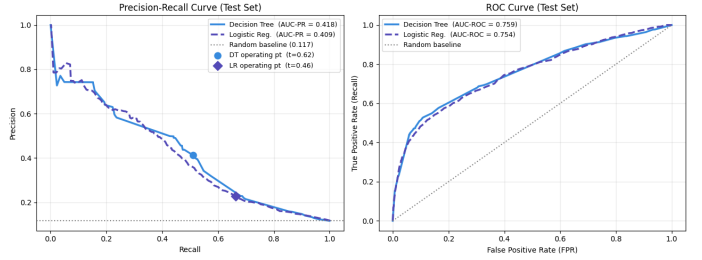


Fig. 3. Precision-Recall and ROC curves

positives compared with 539 for the Decision Tree. However, it also generated substantially more false positives, with 2,376 unnecessary positive predictions compared with only 765 for the Decision Tree. This result reflects the higher Recall but lower Precision behavior of Logistic Regression. In contrast, the Decision Tree produced a more balanced classification outcome with fewer wasted outreach predictions.

Figure 3 compares the Precision-Recall and ROC curves for both models. The Decision Tree achieved slightly higher AUC-PR and AUC-ROC values, indicating stronger overall ranking and classification performance across thresholds. The Precision-Recall curve also shows that the Decision Tree maintains higher Precision across moderate Recall ranges, which is particularly valuable in imbalanced marketing prediction tasks where false positives carry operational costs.

Figure 4 presents the cumulative gains comparison between the two models. The Decision Tree achieved an ALIFT score of approximately 0.729, slightly outperforming Logistic Regression. More importantly, the operating-point comparison demonstrates the practical business tradeoff between the two models. The Decision Tree identified approximately 51% of subscribers while contacting only 14.4% of customers, indicating a highly efficient targeting strategy. Logistic Regression captured approximately 66.4% of subscribers but required contacting 34% of customers, reflecting a broader but less efficient outreach strategy. This comparison illustrates that the choice between models should depend on campaign objectives. If the goal is aggressive lead capture, Logistic Regression may be preferred because of its stronger Recall. If the goal is operational efficiency and reducing unnecessary customer contact, the Decision Tree provides stronger overall targeting performance.

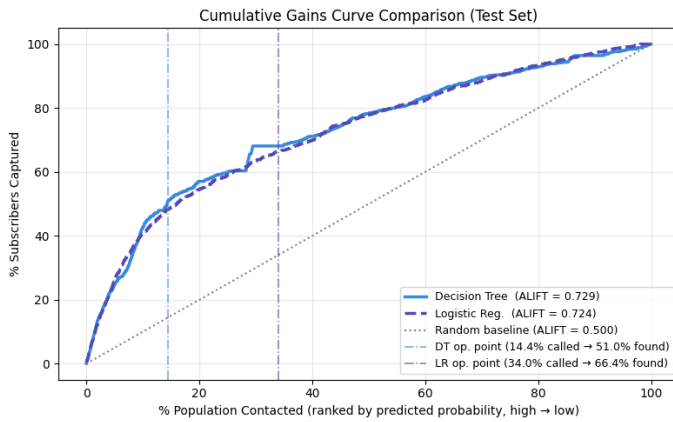


Fig. 4. The cumulative gains comparison between the two models

Several techniques were used to improve generalization and prevent both overfitting and underfitting. For Logistic Regression, L2 regularization penalized excessively large weights and reduces sensitivity to noise in the training data. This helps prevent the model from overfitting while maintaining stable probability estimates. For the Decision Tree, pre-pruning constraints were applied through maximum tree depth and minimum samples per split. These constraints limit unnecessary tree complexity and prevent the model from memorizing small noisy patterns in the training data.

To address the strong class imbalance in the dataset, both models incorporated class weighting. Minority-class subscriber samples received proportionally larger influence during optimization and splitting. This prevents the models from collapsing into majority-class predictions and improves learning performance on actual subscribers. In all, the Decision Tree achieved stronger overall performance than Logistic Regression across most evaluation metrics. On the test set, the Decision Tree achieved higher Precision, Accuracy, F1-score, F2-score, AUC-ROC, AUC-PR, and ALIFT.

Logistic Regression achieved a stronger Recall, indicating that it identified more true subscribers. However, this came at the cost of a substantially larger number of false positives. In a real marketing setting, this would result in significantly more unnecessary customer calls.

The Decision Tree produced a more balanced tradeoff between subscriber identification and operational efficiency. This suggests that nonlinear interactions among campaign outcome, contact information, timing, and financial features are important for predicting customer behavior.

Feature importance analysis showed strong agreement between the two models. Both models identified `poutcome_success` and `contact_unknown` as highly predictive variables. Other important features included `housing`, `month_aug`, `month_jul`, `month_mar`, and `month_nov`. These findings suggest that previous campaign success and communication channel quality are more informative predictors than demographic variables alone.

The lift analysis further demonstrated the effectiveness of

the Decision Tree for marketing prioritization. Customers ranked in the top probability deciles contained a disproportionately large share of actual subscribers, indicating that the model can effectively concentrate marketing resources on the most promising customer groups. Overall, the results show that both models learned meaningful predictive patterns from the dataset, but the Decision Tree provided stronger overall performance for this business objective.

5) Model Deployment & Front-End Application Using Streamlit: We developed a web-based application using Streamlit to support real-time customer subscription prediction for bank marketing campaigns. The app is organized as a multi-page interface, with a home page plus separate prediction pages for Logistic Regression and Decision Tree. This design lets users run single-customer inference quickly without writing code, while keeping the workflow practical for non-technical marketing use.

For model selection, we considered the trade-off between recall and targeting efficiency. Logistic Regression captures more potential subscribers, so it is useful when the campaign goal is aggressive lead capture and missing positives is costly. However, it also creates more false positives, which means more unnecessary outreach. The Decision Tree gives a better overall balance across Precision, F1, F2, AUC-ROC, AUC-PR, and ALIFT, so it is more efficient for operational targeting. Because this project emphasizes both prediction quality and practical marketing efficiency, the Decision Tree is the preferred deployed model, while Logistic Regression remains available when recall-first strategy is needed.

The target population is bank marketing teams, including campaign managers and outbound calling staff. The application benefits them by providing a fast, consistent, and interpretable way to score customers before contact. Instead of broad outreach, teams can prioritize higher-probability customers, which helps reduce wasted calls and improve conversion-focused resource allocation.

The interface layout is form-driven and simple to use. Users input customer and campaign features such as age, loan status, balance, campaign contacts, previous contacts, days since last contact, contact type, month, and previous outcome. Additional fields are available in an advanced section, and users can adjust the decision threshold with a slider. After pressing the Predict button, the app displays the predicted probability, the binary decision at the selected threshold, and a short interpretation message.

The front-end and back-end are connected through Streamlit's event-driven execution flow. When the user presses Predict, the input values are passed to backend Python logic, where model artifacts are loaded from saved files and the same preprocessing rules used in training are applied. The system then builds the model-ready feature vector, runs probability inference, applies threshold logic, and sends results back to the page immediately. This keeps the deployed behavior aligned with offline training and makes the prediction pipeline reliable in practical use.

Fig. 5. Front-end Interface

Fig. 6. Front-end Interface

IV. CONCLUSION

This project developed an end-to-end machine learning pipeline to predict whether bank customers would subscribe to a term deposit using demographic, financial, and campaign-related features. Using the UCI Bank Marketing dataset, we implemented two supervised classification models entirely from scratch with NumPy, Logistic Regression and Decision Tree. The pipeline included data exploration, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation under a highly imbalanced classification setting.

The experimental results show that both models learned meaningful predictive patterns from customer behavior and campaign history, but they exhibited different strengths. Logistic Regression achieved a stronger Recall, meaning it identified a larger proportion of potential subscribers. This makes it more suitable in situations where the bank prioritizes maximizing lead coverage and is willing to tolerate a larger number of unnecessary calls. In contrast, the Decision Tree achieved

stronger overall performance across Precision, F1-score, F2-score, AUC-ROC, AUC-PR, and ALIFT. The model produced a more balanced tradeoff between capturing subscribers and controlling operational cost, making it more practical for real-world campaign targeting.

The comparison between the two models highlights an important business insight for bank marketing prediction systems: model selection should depend on campaign objectives rather than a single evaluation metric. If the primary goal is aggressive customer acquisition, a higher-recall model such as Logistic Regression may be preferred. However, if the bank aims to optimize outreach efficiency and reduce wasted resources, the Decision Tree provides stronger ranking and classification performance. Both models consistently identified previous campaign success and customer contact information as the strongest predictors of subscription behavior. This suggests that behavioral history and communication accessibility are more informative than demographic variables alone. In practical deployment, the models can help banks prioritize high-probability customers before outreach begins, enabling marketing teams to improve conversion efficiency and reduce unnecessary customer contact.

Future improvements could include implementing ensemble methods such as Random Forest or Gradient Boosting, introducing cost-sensitive threshold optimization based on actual campaign economics, and applying probability calibration techniques to improve ranking reliability. Additional fairness analysis across demographic subgroups would also be important before real-world deployment to ensure that automated targeting decisions remain transparent and responsible.

V. TEAM MEMBER CONTRIBUTION

Each team member contributed to both the technical development and the written components of the project. Responsibilities were divided to ensure coverage of the full end-to-end machine learning pipeline, including data processing, model implementation, evaluation, and deployment.

A. Technical Components

Member 1: Zijng Ding (Foundation & Framework)

Zijng Ding was responsible for establishing the overall structure of the project, including defining the problem scope and framing the machine learning pipeline. This included drafting the project title, abstract, and introduction, and identifying the broader impact of the application. The contribution also involved outlining how the proposed system improves efficiency and ethical targeting in financial marketing.

Member 2: Yufan Wang (Literature, Context & Decision Tree Modeling)

Yufan Wang conducted the literature review and background research. This included analyzing prior work on the Bank Marketing dataset and identifying key insights from Moro et al. (2014). Yufan also managed citation formatting and BibTeX references in Overleaf to ensure all sources were correctly integrated into the report. In addition, Yufan contributed to the offline modeling component by implementing and evaluating the Decision Tree classifier, including

model training, hyperparameter tuning, and comparison with Logistic Regression.

Member 3: Yongqi Liu (Data Engineering & Exploration) Yongqi Liu handled data-related components, including dataset understanding, exploratory data analysis, and preprocessing design. This included proposing and implementing visualizations such as histograms, boxplots, and correlation heatmaps, as well as designing preprocessing steps such as one-hot encoding, normalization, and outlier handling.

Member 4: Xinyi Yang (Model Architecture) Xinyi Yang was responsible for implementing and documenting the Logistic Regression model. This included deriving and presenting the mathematical formulation, implementing the algorithm from scratch using NumPy, and defining model inputs and outputs. The contribution also involved justifying why Logistic Regression is appropriate for binary classification tasks in this context.

Member 5: Jade Yang (Evaluation & Risk Analysis) Jade Yang focused on model evaluation and system robustness. This included defining evaluation metrics such as Accuracy, Precision, Recall, and F1-score, along with their mathematical formulations. Jade also designed the train/test split strategy, planned hyperparameter tuning, and identified key risks such as class imbalance and overfitting, along with mitigation strategies.

Member 6: Camile Tong (Deployment & UI/UX Design) Camile Tong was responsible for designing the deployment pipeline and user interface. This included defining the Streamlit application structure, designing input widgets and output displays, and specifying how the front-end connects to the back-end model. The contribution also included outlining expected outcomes and ensuring the system is usable by non-technical users such as bank marketing managers.

B. Writing Components

All team members contributed to writing, reviewing, and refining the final report. Responsibilities were distributed as follows:

- Zijing Ding: Title, Abstract, and Introduction
- Yufan Wang: Background and literature review
- Yongqi Liu: Data Collection, Exploration, and Processing
- Xinyi Yang: Methods and Model Training
- Jade Yang: Model Evaluation
- Camile Tong: Model Deployment, Results

All members participated in editing for clarity, consistency, and formatting to ensure alignment with IEEE standards.

VI. CODE, DOCUMENTATION, AND DEPLOYMENT

Github Repository: <https://github.com/Cassieliu77/Bank-Deposit-Prediction>

Streamlit frontend interface url: <https://bdp-home.streamlit.app/>

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